Timeline of Research on Extracellular ATP-Induced Cell Death and the Evolution of the AICD Concept (1929–2026)

I. Foundations of ATP biology and signaling (1929–1972)

II. Discovery of extracellular ATP-induced cytotoxicity and programmed cell death (1975–1991)

III. P2X7 receptor and inflammasome mechanisms (1996–2009)

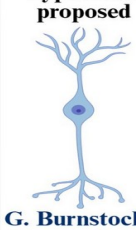
IV. Expansion of ATP-induced cell death mechanisms and cancer therapeutic relevance (2013–2022)

V. Conceptualization of AICD and proposal of AICS (2023–2026)

3

1972

Purinergic neurotransmission hypothesis proposed

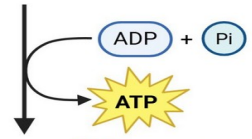


G. Burnstock
Pharmacol Rev.
1972;24(3):509–581.
PMID: 4404211

2

1941

Concept of high-energy phosphate bonds and ATP-mediated energy transfer proposed

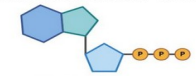


F. Lipmann
Adv Enzymol Relat Areas Mol Biol.
1941;1:99–162.
DOI: 10.1002/9780470122464.ch4

1

1929

Independent discovery and initial chemical characterization of ATP

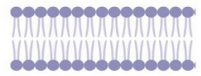


K. Lohmann, C. H. Fiske and Y. Subbarow
Lohmann K.
Naturwissenschaften.
1929;17:624–625.
DOI: 10.1007/BF01506215.
Fiske CH, Subbarow Y.
Science. 1929;70(1816):381–382.DOI: 10.1126/science.70.1816.381.b

4

1975

Extracellular ATP selectively increases membrane permeability in transformed 3T3 fibroblasts



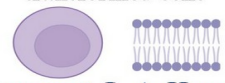
E. Rozengurt,L. A. Heppel

Biochem Biophys Res Commun. 1975;67:1581–1588.
DOI: 10.1016/0006-291X(75)90207-7

5

1977

Extracellular ATP induces extensive and reversible membrane permeabilization in transformed cells



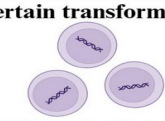
E. Rozengurt, L. A. Heppel, I. Friedberg

J Biol Chem.
1977;252(13):4584–4590.
PMID: 17602
DOI: 10.1016/s0021-9258(17)40202-x

6

1988

Extracellular ATP induces plasma membrane permeabilization and lysis in certain transformed cells



T. Kitagawa,F. Amano, Y. Akamatsu

Biochim. Biophys. Acta.
1988;941:257–263.
DOI:10.1016/0005-2736(88)90186-1

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1991

Extracellular ATP is shown to induce apoptosis



L. M. Zheng,A. Zychlinsky, C.-C. Liu,D. M. Ojcius, J. D.-E. Young

J. Cell Biol. 1991;112(2):279–288.
DOI: 10.1083/jcb.112.2.279

11

2013-2022

Expansion of ATP-induced cell death mechanisms and cancer translation



Studies of multiple ATP-induced cell death modalities, including apoptosis, autophagy-associated cell death, necroptosis, and pyroptosis, expanded toward cancer signaling, prognosis, and therapeutic applications.

10

2009

Tumor cell-released ATP links chemotherapy-induced cell death with antitumor immunity



F. Ghiringhelli,L. Apetoh, A. Tesniere,L. Aymeric, Y. Ma,C. Ortiz et al.

Nat. Med. 2009;
15(10):1170–1178.
DOI:10.1038/nm.2028

9

2006

ATP-induced P2X7-dependent NLRP3 inflammasome activation elucidated



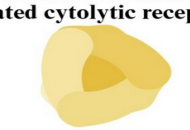
S. Mariathasan,D. S. Weiss, K. Newton,J. McBrideK, O'Rourke et al.

Nature. 2006;
440(7081):228–232.
DOI:10.1038/nature04515

8

1996

P2X7 receptor identified as an ATP-gated cytolytic receptor



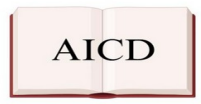
A. Surprenant,F. Rassendren, E. Kawashima,R. A. North, G. Buell

Science. 1996;
272(5262):735–738.
DOI:10.1126/science.272.5262.735

12

2023.11.22

ATP-induced cell death (AICD) concept proposed and defined



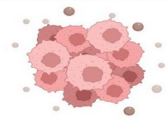
Hao-Ling Zhang et al.

Int J Biol Life Sci.
2023;4(1):15–16.
DOI:10.54097/ijbls.v4i1.04

13

2023.12.24

Systematic framework of AICD mechanisms and cancer translational applications was established



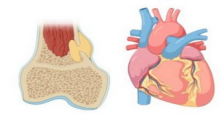
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World J Clin Oncol.
2023;14(12):549-569.
DOI: 10.5306/wjco.v14.i12.549
PMID: 38179405

14

2024–2026

AICD in disease models and cross-disease expansion



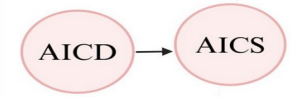
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Front Cell Dev Biol.
2023;11:1324213.
DOI: 10.3389/fcell.2023.1324213
PMID: 38161333
J Inflamm Res. 2026;19:584582.
DOI:10.2147/JIR.S584582.
PMID: 41847423

15

2026

Proposal of the AICS continuum framework



Hao-Ling Zhang et al.

AICD was further conceptualized within the ATP-induced cell state continuum framework.